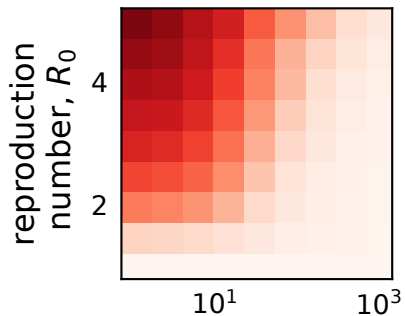
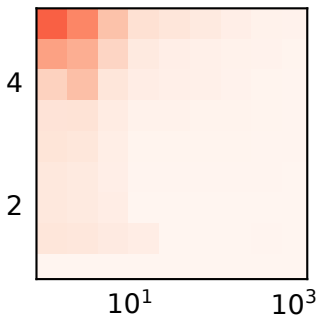


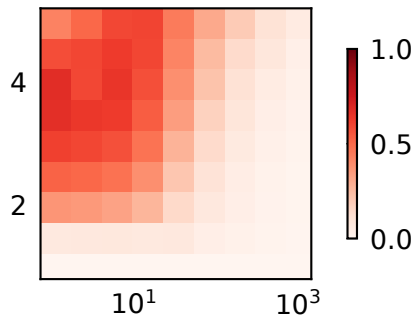
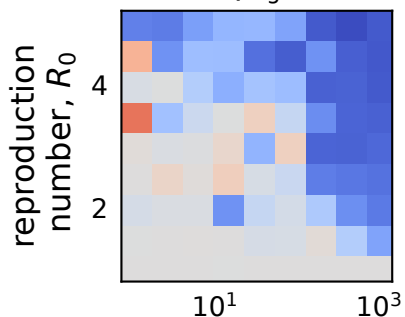
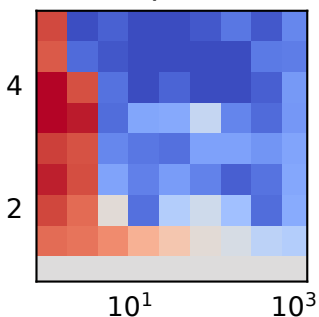
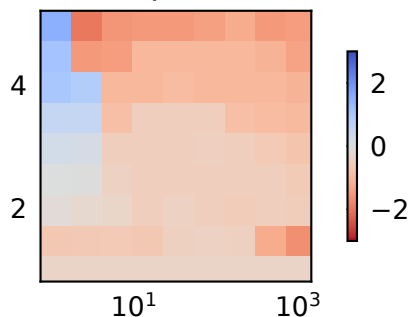
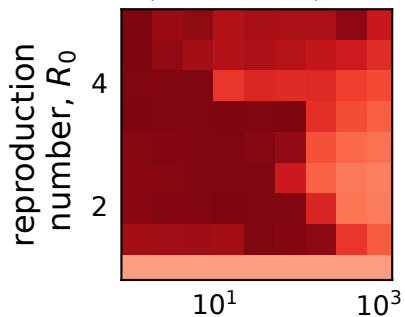
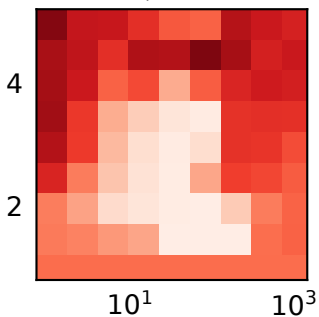
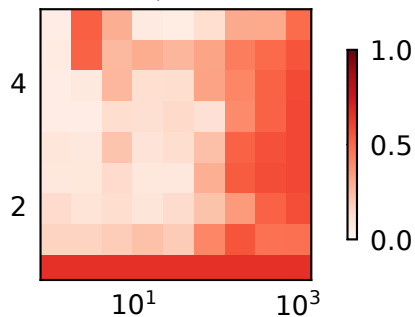
total harm



infection harm



toxicity

 ψ_{Ag}  ψ_{inf}  ψ_{res}  $g_{0,N \rightarrow N^*} = g_{0,E \rightarrow 2E}$  $g_{0,N^* \rightarrow E}$  $g_{0,E \rightarrow \emptyset}$ immunogenicity, κ immunogenicity, κ immunogenicity, κ